

Punith Anumala

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SUMMARY

6+Years' Experience in Design, build, and maintain scalable data pipelines to support data integration, ETL processes, and data warehousing and develop and optimize SQL queries to extract, transform and Proven track record of leveraging Docker for containerized development and deploying scalable, fault-tolerant solutions using Kubernetes.

EXPERIENCE

*Eastern Bank, **Azure Data Engineer***

Feb 2025 - Present

Responsibilities:

- Analysed and developed a modern data solution with Azure PaaS services to enable data visualization, understanding the application's current state and impact on existing business processes.
- Implemented Azure Data Factory (ADF) extensively for ingesting data from various source systems, including relational and unstructured data, to meet business functional requirements.
- Used Spark to perform transformations and actions, saving the resulting data to HDFS, which is then moved to the target database, Snowflake and Leveraged Kafka as a messaging system to implement real-time Streaming solutions using Spark Streaming for efficient data processing and Worked on Big Data Integration & Analytics, utilizing Hadoop, SOLR, Pyspark, Kafka, Storm, and Web Methods in the analysis, design, and execution of Hadoop and Big Data projects.
- Expertise in Business Intelligence and Data Visualization tools like Tableau, connecting to various data sources and building informative graphs and dashboards and designed and implemented Infrastructure as Code using Terraform, enabling the automated provisioning and scaling of Azure resources.
- Developed and managed Azure Cosmos DB & Neo4j for scalable, globally distributed, low-latency data storage and implemented advanced Azure Machine Learning solutions, integrating models into existing pipelines for enhanced predictive analytics.
- Ensured data integrity and consistency during migration by resolving compatibility issues with T-SQL scripting.
- Implemented Synapse Integration with Azure Data bricks notebooks, reducing development work by half and improving performance by implementing dynamic partition switches for Synapse loading optimization and Implemented CI/CD pipelines using Jenkins to automate the build, test, and deployment processes, ensuring faster and more reliable delivery
- Implemented Docker and Kubernetes to containerize and orchestrate data pipelines, ensuring scalable, consistent, and efficient deployment of applications across multiple environments, improving system reliability and operational efficiency.
- Utilized GIT for version control, Maven for project build management, and ETL tools like Informatica and Talend for data extraction, transformation, and loading, ensuring streamlined development and data integration processes.

*The Hanover Insurance Group, **AWS Data Engineer***

Feb 2024 - Dec 2024

Responsibilities:

- Developed, deployed, and managed scalable data pipelines using AWS services like AWS Glue, AWS Lambda, and Amazon Kinesis to handle data ingestion, transformation, and loading for UMass Auxiliary Enterprises.
- Utilized Amazon S3 to build and manage data lakes for storing structured and unstructured data, providing scalable and secure data storage solutions and Integrated AWS Dynamo DB with AWS Lambda to store and backup values of items in real-time, ensuring data consistency and availability. Loaded data from BDW Oracle and Teradata into HDFS using Sqoop and implemented interactive web screens with AJAX, JSON, and JavaScript to enhance user experience.
- Collaborated with a cross-functional team to migrate on-prem Hadoop/Hive pipelines to AWS, leveraging Amazon EMR (Spark), Glue, and S3 to modernize data workflows—achieving a 40% reduction in operational overhead and significantly enhancing data quality.
- Engineered a scalable architecture for a secure, highly available data lake on Amazon S3 with intelligent tiering, KMS encryption, and IAM-based data access controls, storing 50M+ structured and semi-structured records.
- Built event-driven ETL pipelines with Spark, Flink, and Glue, orchestrated via Airflow and Step Functions, reducing batch latency from 2 hrs to 75 mins and enhancing data reporting for portfolio performance.
- Provisioned infrastructure using Terraform (IaC), enabling repeatable, auditable deployments of EC2, EMR, RDS, and Redshift for QA and prod environments, supporting robust quality assurance and data controls.
- Integrated XML/JSON feeds from REST APIs using API Gateway and Lambda; loaded enriched data into Redshift, RDS, and DynamoDB, facilitating low-latency data access and enhancing master data management for analysts.
- Delivered real-time data streaming with Kinesis, Flink, and AWS Data Streams to ingest and process market tick data within 5 seconds, supporting advanced data mining and quantitative trading models.
- Architected cloud-native data solutions using Docker for modular ETL workflows; implemented auto-scaling and EMR tuning strategies, reducing AWS compute costs by 25% and boosting performance management.

- Mentored engineers on PySpark, Airflow DAGs, and modular design, raising code quality KPIs by 30% while maintaining data governance, quality control, and security compliance.
- Managed CI/CD pipelines in GitLab with Git version control systems, driving collaborative development and enabling continuous delivery of secure, production-grade data strategy and ingestion pipelines.
- Ensured data quality, integrity, and end-to-end testing while documenting existing programs and codes to improve future enhancements and troubleshooting processes and Integrated Snowflake for efficient data storage and processing, incorporating machine learning models to analyse large datasets and provide predictive insights for operational optimization.
- Leveraged Tableau and Power BI for data visualization, creating interactive dashboards and reports to present operational metrics and performance insights to UMass Auxiliary Enterprises stakeholders.
- Applied Apache Airflow for automating and scheduling data workflows, ensuring seamless integration of data pipelines and optimizing the performance of ETL processes across various systems.

VOZIQ AI, Data Engineer

Jun 2020 - Dec 2022

Responsibilities:

- Designed and implemented Jenkins jobs to automate continuous integration and CI/CD pipeline processes, ensuring streamlined development workflows at Standard Chartered Bank.
- Developed and maintained scalable data pipelines using GCP services like Dataflow, Apache Beam, and Cloud Pub/Sub for efficient streaming and batch processing of financial data.
- Assisted in migrating on-prem Hadoop systems to GCP (Google Cloud Platform), ensuring seamless integration with cloud services and improving operational efficiency for financial data management and worked extensively with Pyspark to create data frames and perform complex transformations on large financial datasets, optimizing data flow processes.
- Built and maintained scalable data lakes on Amazon S3 with data partitioning, Glue Data Catalog, and metadata management, supporting data discoverability, governance, and efficient processing of structured/unstructured data.
- Designed and implemented data marts using dimensional modeling (star/snowflake schemas) aligned with data mesh principles to support decentralized data ownership and scalable BI solutions.
- Built and optimized cloud data warehousing solutions with Redshift and Redshift Spectrum, applying query optimization techniques to accelerate analytics performance and support Amazon QuickSight integration.
- Implemented root cause analysis, troubleshooting and pipeline debugging for ETL workflows and Spark processes, resolving data quality issues and improving reliability and processing speed by 30%.
- Developed monitoring/alerting with CloudWatch, Grafana, and Datadog, enabling observability and user acceptance testing for validating data pipeline performance, ensuring SLA compliance.
- Utilized AWS/GCP cloud services like Cloud Storage, DataProc, Big Query, and S3 for data processing and storage, enabling fast and reliable data handling for bank operations and Integrated Data Lake, SQL Database, and Logic Apps to build an efficient data architecture that supports analytics and business intelligence across the organization.
- Developed multi-cloud strategies by leveraging GCP PAAS offerings and AWS to migrate legacy financial systems, optimizing performance and cost-efficiency across multiple cloud environments.
- Implemented Apache Airflow for orchestration of complex workflows, automating data ingestion, transformation, and reporting processes to improve operational efficiency and Utilized Docker and Kubernetes to containerize and orchestrate applications, enhancing the scalability and deployment efficiency of financial analytics
- Leveraged Terraform to automate the provisioning and management of cloud infrastructure on GCP.

ICICI Prudential Life Insurance, Data Engineer

Jun 2018 - May 2020

Responsibilities:

- Built scalable data pipelines using Python, Pyspark, Hive SQL, and Presto to extract, transform, and load data from relational and non-relational databases such as Cassandra and HDFS, ensuring efficient data transformation and cleansing.
- Expert in creating ETL jobs with Hive SQL, Presto SQL, and Spark SQL to automate data ingestion, transformation, and validation processes, improving operational efficiency for insurance data analysis.
- Developed dashboards in Tableau by integrating data from sources like Big Query and Presto SQL through ODBC connections, providing visual insights into policyholder behaviour and claim trends.
- Utilized SAP SD Module for transaction processing, generating reports related to customer handling and sales, and integrated SQL Server with FTP processes for automated data fetching and table updates.
- Managed large-scale datasets using Pandas and SQL, ensuring data accuracy and optimized data flow between sources and insurance databases, supporting comprehensive reporting and analytics.
- Implemented continuous integration and continuous delivery (CI/CD) pipelines using Docker, Kubernetes, and Jenkins for seamless deployment of custom applications and efficient cloud management.
- Designed and maintained infrastructure as code using AWS Terraform templates, enabling automated provisioning and scaling of cloud resources, ensuring flexibility and scalability in insurance data processing.
- Developed and optimized real-time data streaming solutions using Apache Kafka and Spark Streaming for processing live insurance transaction data, enabling faster claim processing and real-time analytics.

- Integrated machine learning models using Python and scikit-learn to predict customer churn and claims forecasting, enhancing decision-making in underwriting and risk assessment and Worked with RESTful APIs and Flask to integrate third-party insurance data sources, improving data sharing and streamlining policyholder data synchronization across platforms.
- Designed and deployed data warehouses using Snowflake and Google Big Query, supporting scalable analytics and enhancing reporting capabilities for business intelligence teams.

EDUCATION

Clark University

Master • IT Project Management

MLRIT

Computer Science

LICENSES & CERTIFICATIONS

AWS Certified Data engineer – Associate

Microsoft Power BI - Data Analyst

AWS Cloud Practitioner - Foundational

SKILLS

Python • NumPy • Pandas • PyTorch • Java • Scala • R • Bash • Hadoop • Hive